

Propagation and Antenna

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Notes

- Our subject code is EX 653.
- This document incorporates for both new and old question paper, both being EX 653.
- If a question was asked in Regular Exam, it is kept in **boldened** form. If it was asked in back exam, it is in regular font.
- Months are marked as:
 - Ba: Baisakh
 - Jth: Jestha
 - Asa: Ashar
 - Shr: Shrawan
 - Bh: Bhadra
 - Ash: Ashwin
 - Ka: Kartik
 - Mng: Mangsir
 - Po: Poush
 - Ma: Magh
 - Ch: Chaitra

Contents

1	Introduction	4
1.1	Retarded Potentials: EM wave generation	4
1.1.1	Conduction current	4
1.1.2	Short uniform current dipole	4
1.1.3	Radiated electric and magnetic fields	4
1.2	Short uniform current dipole, short dipole and long dipole	4
1.3	Radiation patterns	4
1.4	Input impedance	4
1.5	Introduction to Antenna	4
1.6	Antenna theorems	5
1.6.1	Reciprocity & Superposition	5
1.6.2	Thevenin & Maximum power transfer Theorem	5
1.6.3	Compensation	5
1.6.4	Equality of directional patterns	5
1.6.5	Equivalence of receiving and transmitting impedances	5
2	Antenna Parameters and Arrays	6
2.1	Basic Antenna Parameters	6
2.2	Pattern multiplication	6
2.2.1	Linear & 2-D antenna arrays	7
2.2.2	End fire	7
2.2.3	Broadside arrays	7
3	Antennas Classification	8
3.1	Broad Classification	8
3.1.1	Isotropic Antenna	8
3.1.2	Directional Antenna	8
3.1.3	Omnidirectional Antenna	8
3.2	Linear Dipole/Monopole	8
3.2.1	Short Dipole	8
3.2.2	Half-Wave Dipole	8
3.2.3	Folded Dipole	8
3.2.4	Monopole	8
3.3	Travelling Wave Antenna	8
3.3.1	Single Wire	8
3.3.2	Vee Antenna	9
3.3.3	Rhombic Antenna	9
3.4	Aperture Antennas	9
3.4.1	Horn Antenna	9
3.4.2	Slot Antenna	10
3.4.3	Reflector Antenna	10
3.5	Array Antenna	10
3.5.1	Yagi-Uda	10
3.5.2	Log-Periodic	11
3.5.3	Phased Array	11
3.6	Broadband & Special Elements	11
3.6.1	Helical Antenna	11
3.6.2	Microstrip Antenna	11

3.6.3	Spiral Antenna	11
3.6.4	Loop Antenna	11
3.7	Other	11
3.7.1	Large Plane Sheet	11
3.7.2	Small Plane Sheet	11
4	Propagation and Radio Frequency Spectrum	12
4.1	Ground or Surface Wave	12
4.2	Space wave; direct and ground reflected wave, duct propagation	12
4.3	Ionospheric or sky wave; critical frequency, MUF, Skip distance	12
4.4	Tropospheric wave	13
4.5	RF spectrum and its propagation characteristics	13
4.6	Numerical	14
5	Propagation between Antennas	15
5.1	Free Space propagation	15
5.1.1	Power density of receiving antenna	15
5.1.2	Path loss	15
5.2	Plane earth propagation	15
5.2.1	Ground Reflection	15
5.2.2	Effective antenna heights	15
5.2.3	The Two Ray	15
5.3	Propagation Model, Path Loss	15
5.4	Fresnel Zones and Knife edge diffraction	15
5.5	Numerical	16
6	Optical fibres	17
6.1	Optical Fibre Communication System	17
6.1.1	Merits/Demerits	17
6.2	Types of optical fibre and its Structural differences	17
6.3	Light propagation characteristics	18
6.4	NA in optical fibre	18
6.5	Losses	18
6.6	Light Source and Photo Detector	18
7	Don't know where this fits	18

1 Introduction

(6 Hours / 10-14 Marks)

1.1 Retarded Potentials: EM wave generation

1. What is retarded potential? [2] (76 Bh, 72 Ash, 81 Ash)
|→ and state its importance. [2] (74 Bh)
2. What is retarded time? [2] (82 Ka, 80 Ash)
3. Write the expression of retarded scalar potential and vector potential of infinitesimal dipole antenna in frauhofer for field region. [4] (72 Ash)
|→ Explain retarded vector and scalar potential with mathematical expression. [5] (80 Ash)

1.1.1 Conduction current

1. Explain how electromagnetic waves are generated by a conductor. [4] (76 Bh)

1.1.2 Short uniform current dipole

1. Explain the operation of infinitesimal dipole with the help of mathematical relations and the field pattern. [4] (71 Ma) [6] (80 Ch, 74 Ba) [8] (70 Bh)

1.1.3 Radiated electric and magnetic fields

1.2 Short uniform current dipole, short dipole and long dipole

1. Find out the retarded vector potential of an electric current at a distance, r from a point source for the far field region. [4] (82 Ka)
2. During the transmission mode, explain the mechanism involved by which the electric lines of force are detached from the dipole antenna to form free space waves. [4] (70 Ma)
3. Explain how single wire can radiate electromagnetic waves with necessary radiation patterns and conditions. [4] (79 Ash)

1.3 Radiation patterns

1.4 Input impedance

1.5 Introduction to Antenna

1. Define antenna. [2] (81 Ch, 80 Ch, 79 Ch, 78 Ch, 75 Bh, 70 Bh, 75 Ba, 73 Ma)
2. Explain the conditions that any type of conductor can radiate electromagnetic wave or not. [4] (75 Bh)
|→ Explain how EM waves are generated by a conductor. [4] (75 Ba)

1.6 Antenna theorems

1.6.1 Reciprocity & Superposition

1. State the following antenna theorem:

|→ Reciprocity. [2] (**77 Ch,75 Bh,74 Bh**, 79 Ash, 73 Ma, 72 Ma) [3] (**79 Ch,78 Ch**, 81 Ash)
|→ Superposition theorem. [2] (72 Ma) [3] (**76 Bh**)

1.6.2 Thevenin & Maximum power transfer Theorem

1. Explain the following antenna theorems: Maximum Power Transfer Theorem.

[2] (**77 Ch**) [3] (**80 Ch,76 Bh**, 82 Ka)

2. State the following theorem: Thevenin.

[3] (**79 Ch**, 81 Ash)

3. Explain superposition theorem and maximum power transfer theorem with diagram. [3+3] (80 Ash)

1.6.3 Compensation

1. Write short notes on: Compensation theorem. [2] (**78 Ch,75 Bh,74 Bh**, 73 Ma) [3] (**81 Ch,80 Ch**)

1.6.4 Equality of directional patterns

1. State equality directional patterns theorem.

[2] (79 Ash)

1.6.5 Equivalence of receiving and transmitting impedances

2 Antenna Parameters and Arrays

(6 Hours / 4-8 Marks)

2.1 Basic Antenna Parameters

1. Explain the following antenna parameters/short notes:
 - a. Receiving antenna gain (G_r) [1.5] (76 Ba) [3] (**81 Ch**)
 - b. Free space loss (L_{FS}) [3] (**81 Ch**)
 - c. Directivity. [1.5] (76 Ba) [2] (70 Ma) [3] (**79 Ch**) [4] (**72 Ash**)
 - d. Polarization. [2] (70 Ma) [3] (**81 Ch, 79 Ch, 79 Ch, 75 Bh, 74 Bh**)
 - e. Linear polarization. [3] (79 Ash, 72 Ma)
 - f. Input impedance. [3] (**79 Ch**)
 - g. Radiation Patterns. [2] (**74 Bh**) [3] (**77 Ch**, 80 Ash, 76 Ba)
 - h. Radiation Pattern Lobes. [2] (70 Ma)
 - i. Beam Width. [3] (**77 Ch**, 76 Ba)
 - j. Beam width with mathematical derivations if necessary. [2] (72 Ma)
 - k. Antenna efficiency. [3] (**75 Bh**) [4] (**72 Ash**)
 - l. Antenna efficiency with mathematical derivations if necessary. [2] (72 Ma)
 - m. Radiation Efficiency. [3] (82 Ka)
 - n. Bandwidth. [3] (**75 Bh**)
 - o. Half power bandwidth. [2] (**70 Bh**)
 - p. Antenna gain. [2] (**74 Bh**)
 - q. Antenna gain, with mathematical derivations if necessary. [2] (72 Ma) [3] (80 Ash)
 - r. Power-flux density, with mathematical derivation if necessary. [3] (80 Ash)
 - s. Write short notes on: Power flux density (PFD). [3] (**80 Ch**)
 - t. Effective Isotropic Radiated Power (EIRP). [3] (82 Ka, 80 Ash, 72 Ma) [4] (**77 Ch**, 73 Ma)
 - u. Receiving power entity (P_r) [3] (**81 Ash**)
2. Why antenna radiation pattern lobes are important during antenna design? [3] (71 Ma)
3. Explain the different conditions for a simple conductor to be a good antenna. [4] (**79 Ch**)
4. Sketch current distribution and radiation pattern of center-fed vertical dipole for: [4] (76 Ba)
 - a. $\lambda/2$
 - b. $3\lambda/2$
 |→ for (a), (b) and (c): 2λ [6] (72 Ma)
5. Differentiate between: $\frac{\lambda}{2}, \lambda, 3\frac{\lambda}{2}$ length dipoles in terms of their individual radiation pattern, self impedance and directivity. [6] (70 Ma)
6. Describe the antenna gain, antenna efficiency and directivity of antenna with mathematical derivation if necessary. [6] (75 Ba)
7. How can we increase the directivity of an antenna? [2] (79 Ash)
8. List the parameters of antenna and explain any three of them. [6] (73 Ma)
 - |→ What are basic antenna parameters? Explain briefly on any 4. [4] (71 Ma)
 - |→ Explain any 5 parameters of antenna. [5] (**71 Bh**)

2.2 Pattern multiplication

1. Explain the principle of pattern multiplication with a suitable example. [3] (**81 Ash**)

2.2.1 Linear & 2-D antenna arrays

1. What is the importance of antenna array? [2] (76 Ba)
2. Why we need antenna array rather than changing the length of single antenna? [2] (72 Ma)
3. State the principle of pattern multiplication. [2] (**70 Bh**)
4. Define pattern multiplication. [2] (80 Ash)
5. What is the importance of antenna array? [2] (**77 Ch**)
6. What is a linear array? [1] (**73 Bh**)

2.2.2 End fire

1. Write the features of end-fire array. [4] (**78 Ch**)
2. Describe the principle of End fire and broadside array. [6] (**77 Ch**)
3. Differentiate between a broadside array antenna and end-fire array antenna. [3] (81 Ash)
4. Draw the resultant pattern for an End Fire Array with dipoles perpendicular to the axis of array with $d = \frac{\lambda}{2}$ & $\alpha = \pi$. [4] (70 Ma)
5. Derive the expressions for the total field in case of two isotropic point sources with
 - a. an equal amplitude and opposite phase. [3+3] (72 Ma) [4+4] (**81 Ch**)
 - b. an equal amplitude and same phase.
|→ Derive a relation for the field intensity for the array of two element isotropic radiators and current equal in magnitude but opposite in phase. Show the condition for maxima and minima direction of intensity and draw the radiation pattern. [6] (**75 Bh**) [6+3] (76 Ba)
|→ Derive a relation for the field intensity for the array of two elements isotropic radiators in various conditions. [5] (**71 Bh**)
6. Derive a mathematical relation to calculate the field intensity of an array for two element isotropic radiators. [4] (70 Ma)
7. Use the principle to obtain a wave pattern for array of two short dipoles for following cases:
 - a. Dipoles aligned perpendicular to the array axis with $d = \lambda/2, \alpha = 0$
 - b. Dipoles aligned perpendicular to the array axis with $d = \lambda/2, \alpha = \pi$ [3+3] (**70 Bh**)
8. Derive the expression for far field pattern of an array of two isotropic point sources having equal amplitude and quadrature phase with $d = \lambda/4$. [6] (80 Ash)
9. Derive the expression for the total field in case of two isotropic point sources with an equal amplitude and quadrature phase and draw the radiation field patterns when spacing between the two point sources are $\lambda/2$ and $\lambda/4$. [5] (82 Ka)
10. Derive the relation for the field intensity of linear array of 2 isotropic radiators with equal potential and 180° out of phase. [3] (71 Ma)

2.2.3 Broadside arrays

1. Differentiate between End fire and broadside array. [4] (**73 Bh**) [6] (76 Ba)
|→ same, but in “write short note” question. [4] (75 Ba)
2. Show the condition for broad side and end fire array with necessary diagrams. [6] (**71 Bh**)

3 Antennas Classification

(10 Hours / Marks)

3.1 Broad Classification

3.1.1 Isotropic Antenna

3.1.2 Directional Antenna

3.1.3 Omnidirectional Antenna

1. List the characteristics of onmidirectional antenna. [2] (72 Ash)
2. Sketch the current distribution and radiation pattern of folded dipole antenna. [3] (81 Ch)
3. Sketch the current distribution and radiation pattern of the infinitesimal dipole antenna. [3] (82 Ka)

3.2 Linear Dipole/Monopole

3.2.1 Short Dipole

3.2.2 Half-Wave Dipole

3.2.3 Folded Dipole

3.2.4 Monopole

1. Write short notes on: Monopole antenna. [3] (80 Ash)
2. Write short notes on: Marconi antenna. [3] (79 Ash) [4] (77 Ch,75 Bh,72 Ash, 73 Ma)
3. Explain the operation of $\lambda/4$ monopole antenna with the help of necessary diagram and field pattern. [6] (72 Ash)
4. List the major characteristics of Marconi antenna with necessary figures. [6] (75 Ba)
5. Explain the working principle and design of Marconni antenna. [5] (71 Bh)

3.3 Travelling Wave Antenna

1. What is travelling wave antenna? [1] (71 Ma) [2] (72 Ma)
2. List the various types of wired antenna with their radiation pattern and polarization. [4] (73 Ma)
3. What is the main difference between standing wave antenna and traveling wave antenna? [2] (71 Ma) [3] (73 Ma)

3.3.1 Single Wire

1. Explain the radiation mechanism and conditions of a single wire antenna that can radiate electromagnetic waves or not. [4] (81 Ch)
|→ Explain the radiation mechanism in single wire antenna. [3] (78 Ch)
|→ How does a single wire act as an antenna? [2] (70 Ma)
|→ Explain how single wire can radiate EM waves with necessary radiation patterns and polarization. [4] (72 Ash)

2. Explain briefly radiation mechanism in single wire antenna. [5] (**73 Bh**)
3. Compare resonant and non-resonant antennas. [2] (82 Ka)
4. Explain traveling wave long wire antenna with required diagrams. [3] (**79 Ch**)

3.3.2 Vee Antenna

1. Explain working principle of a V-antenna with
 - |→structure, radiation pattern. [6] (**77 Ch**)
 - |→construction, adv and application. [6] (**76 Bh**)
 - |→construction, characteristics and neat sketch. [5] (73 Ma)
 - |→construction, principle and feature. [6] (82 Ka)
 - |→construction and characteristics. [6] (80 Ash) [7] (71 Ma, 70 Ma)
 - |→characteristics and types. [5] (72 Ma)
2. Sketch the pattern of both types of V antennas. [2] (**76 Bh**)
3. Write short notes on: V antenna. [4] (76 Ba)

3.3.3 Rhombic Antenna

1. Explain the working principle, characteristics of rhombic antenna. [4] (**70 Bh**)
 - |→with advantage and applications. [7] (**79 Ch**)
 - |→with construction. [6] (71 Ma)
 - |→with construction and necessary diagram. [8] (**78 Ch, 73 Bh**)
 - |→with operation. [7] (**72 Ash**)
2. Write short notes on: Rhombic Antenna. [3] (**76 Bh**)
3. Explain the working principle and design of Rhombic antenna. [5] (**71 Bh**)

3.4 Aperture Antennas

1. Write short notes: Aperture Antenna. [3] (**81 Ch**)
 - |→What is aperture antenna. [1] (**75 Bh**) [2] (81 Ash)
2. What are the advantages of aperture antenna? [2] (71 Ma)
3. How are aperture type of antennas different from the conventional antennas? [2] (**77 Ch**)

3.4.1 Horn Antenna

1. Write short notes: Horn Antenna. [3] (**79 Ch**, 72 Ma)
2. Write short notes: Pyramidal horn antenna. [3] (**73 Bh**)
3. Explain the theory of radiation from a rectangular horn. [4] (**77 Ch**)
4. Derive the relation for flare angle and length of a pyramidal horn antenna. [5] (**70 Bh**)
5. List out the type of horn antenna and explain rectangular horn briefly. [6] (71 Ma)

3.4.2 Slot Antenna

3.4.3 Reflector Antenna

1. What is reflector antenna? [1] (76 Bh)
2. Explain the special features of reflector antenna and discuss on different types of feed used with necessary diagrams. [8] (78 Ch)
3. Write short notes on: Corner reflector antenna. [4] (76 Ba)
4. Parabolic Reflector
 - (a) Explain parabolic reflector:
 - operation. [5] (79 Ch)
 - with diagrams and its feeding systems. [6] (76 Bh)
 - construction, working principle and feeding mechanism. [6] (75 Bh) [8] (73 Bh)
 - (b) Write short notes on: Parabolic reflector antenna. [4] (75 Ba)
 - (c) Explain dual-parabolic reflector antennas with their types, characteristics and application. [6] (81 Ash)
 - (d) Describe the cassegrain method of feeding parabolic reflectors. [6] (73 Ma)
 - (e) What is Cassegrain feed system. [3] (79 Ch)
5. Write short notes on: Cassegrain antenna. [3] (80 Ch, 82 Ka)

3.5 Array Antenna

3.5.1 Yagi-Uda

1. Explain Yag-Uda antenna with construction, working principle:
 - with design, advantage and application. [7] (75 Bh)
 - with design. [5] (73 Ma) [8] (74 Bh)
2. What is the main role of driving element of Yagi-Uda antenna? [3] (80 Ch)
3. Explain significance of parasitic elements in Yagi-Uda array. [2] (80 Ash, 76 Ba, 75 Ba)
4. Name the parasitic elements used in Yagi-Uda array. [1] (80 Ash, 76 Ba, 75 Ba)
5. What are the advantages of Yagi antenna? [2] (72 Ma)
6. Design a 3 element Yagi antenna for the operating frequency band of 2785 MHz to 2895 MHz and dipole as a driven element element. Take an effective length of the dipole $0.48\lambda_c$ and consider spacing is $0.15\lambda_c$. [6] (82 Ka)
7. Design a 3-element Yagi antenna with the operating frequency of 3 GHz and dipole as driven element. [4] (72 Ma)
8. Design 5 element Yagi-Uda antenna for the operating frequency band of 785 MHz to 895 MHz and dipole as a driven element. Take an effective length of the dipole $0.48\lambda_c$. [6] (81 Ch)
9. Design five-element Yagi-Uda antenna with operating frequency 760 MHz and length of driver element $0.48\lambda_c$. [6] (80 Ch)
 - for 100 MHz, effective length = $0.48\lambda_c$ and spacing = $0.15\lambda_c$. [7] (76 Bh)
 - for 800 MHz, effective length = 0.48π spacing. [6] (73 Ma)
 - for 100 MHz, length not given. [6] (72 Ash)

3.5.2 Log-Periodic

1. Explain the construction, working principle of Log-periodic antenna. [8] (70 Ma)
|→with charactersitics. [8] (70 Ma)
|→with characteristics and necessary diagram. [6] (81 Ch)
|→with design. [8] (71 Bh) [10] (80 Bh)
|→with design, adv and appplication. [7] (81 Ash)
|→with design and necessary diagram. [7] (73 Ma)
2. Compare Yagi-Uda antenna with Log periodic dipole array. [2] (71 Bh) [3] (80 Ash) [4] (76 Ba, 75 Ba) [6] (79 Ash)
3. Write short notes on: Logarithmic antenna. [4] (74 Bh)

3.5.3 Phased Array

3.6 Broadband & Special Elements

3.6.1 Helical Antenna

1. Explain the helical antenna with respect to structural design, working principle in axial mode and normal mode, their merits, demerits and applications. [8] (77 Ch)
2. Describe a helical antenna with the help of a neat diagram. [4] (79 Ash)
3. Explain two modes of operation of helical antenna with diagrams. [4] (79 Ash)
4. Write short notes on: Helical Antenna. [3] (73 Bh, 80 Ash) [4] (74 Bh)

3.6.2 Microstrip Antenna

1. Write short notes on: Microstrip antenna. [3] (79 Ch, 78 Ch)
|→Write short notes on: Printed antenna. [3] (73 Bh)

3.6.3 Spiral Antenna

3.6.4 Loop Antenna

3.7 Other

3.7.1 Large Plane Sheet

3.7.2 Small Plane Sheet

4 Propagation and Radio Frequency Spectrum

(6 Hours / Marks)

1. Define transmission loss. [2] (76 Ba)

4.1 Ground or Surface Wave

1. Explain the surface wave propagation. [5] (81 Ch)
2. Write major adv and disadv for surface wave propagation. [4] (81 Ch)
3. Write the characteristics of ground wave propagation. [4] (78 Ch)
4. Write down the factors which affect the surface wave communication. [2] (73 Bh)
5. Write short notes on: Medium wave. [3] (80 Ash)

4.2 Space wave; direct and ground reflected wave, duct propagation

1. What is a tropospheric duct? [2] (81 Ash)
2. Explain the phenomenon of Duct propagation. [4] (70 Ma) [5] (72 Ash, 73 Ma)
|→ in radio wave propagation. [4] (71 Ma)
3. Describe multiple HOP and explain how atmospheric ducts can be used for microwave propagation. [2+6] (79 Ch)
4. Write down the factors which affect the space wave communication. [5] (71 Bh, 79 Ash)
5. Explain the effect of space wave propagation on the ground of plane and actual earth. [5] (70 Bh)
[10] (70 Ma)
6. Write short notes: Effect of earth's curvature in space wave propagation. [3] (79 Ash)

4.3 Ionospheric or sky wave; critical frequency, MUF, Skip distance

1. Write short notes on: Ionospheric wave. [3] (81 Ash)
2. What is the sky wave? [2] (80 Ash)
3. What is MUF? [2] (75 Bh, 74 Bh, 81 Ash)
|→ Define MUF with necessary derivations. [2] (71 M)
4. Explain what will happen if a radio wave with a frequency greater than critical frequency is propagated to the ionosphere? Will it return back? [3] (79 Ash)
5. Describe briefly the structure of ionosphere & its uses for radio frequency spectrum. [4] (78 Ch)
|→ Draw structure of ionosphere & explain the mechanism of ionosphere propagation. [6] (76 Ba)
|→ Explain all layers of ionosphere and their importance to radio wave communication. [6] (75 Bh)
|→ Explain the different layers of ionosphere. [5] (71 Ma)
6. Explain the operation of ionospheric communication with the help of different layers created in the atmosphere. [6] (80 Ash) [8] (77 Ch)
|→ Explain the importance of different layers in ionospheric communication. [6] (76 Bh)

7. Explain the ionospheric wave propagation showing its different layers. [6] (72 Ash)
8. With a mathematical relation of refractive index of ionospheric layer derive a relation of critical frequency and MUF of radio waves with necessary explanation. Consider the earth is not curved. [8] (71 Bh)
9. What are the major characteristics of ionosphere? [3] (71 Ma)
10. Define a critical frequency f_{CRT} . [2] (81 Ch, 70 Ma)
 |→ Write short notes on: Critical frequency. [4] (76 Ba)
 |→ Define critical frequency with necessary derivations. [2] (71 Ma)
11. What is skip distance? [1] (74 Bh) [2] (72 Ma)
12. Explain virtual height of a layer. [2] (74 Bh, 72 Ma, 70 Ma)
13. Define maximum usable frequency. [2] (72 Ma)
14. Derive an expression for the relation between skip distance and maximum usable frequency (f_{MUF}) in SW propagation assuming that the earth is flat. [6] (81 C, 78 Ch, 74 Bh) [7] (76 Bh, 80 Ash)
 |→ assuming earth's curvature. [7] (79 Ch) [8] (75 Bh)
15. Derive the relation between the critical frequency and maximum usable frequency. [4] (72 Ma)
 |→ when earth is assumed its not curved. [5] (79 Ash)
16. Derive relationship between the critical frequency and skip distance considering that the earth is curvature surface. [7] (81 Ash)
17. Briefly discuss the propagation characteristics of space wave and sky wave. [8] (73 Bh)

4.4 Tropospheric wave

1. Find the maximum range of tropospheric transmission for which the transmitting antenna height is 100 ft and receiving antenna is 50 ft. [5] (73 Ma)
2. Write short notes on: Tropospheric scatter propagation. [3] (73 Bh)

4.5 RF spectrum and its propagation characteristics

1. Write short notes on: Radio frequency spectrum. [3] (78 Ch, 72 Ash) [4] (75 Bh)
 |→ Define radio frequency spectrum. [2] (74 Bh)
2. What is radio frequency spectrum? [2] (80 Ch)
3. Write short notes on: Radio frequency spectrum. [3] (81 Ash)
4. Give major propagation characteristics of VHF and UHF bands. [6] (74 Bh)
5. List the major features of UHF wave propagation. [3] (79 Ash)
6. List the main characteristics of UHF band. [4] (81 Ch, 81 Ash)
7. Explain characteristics of MW radio propagation. [3] (73 Bh, 71 Bh) [4] (80 Ch)
8. Write short notes on: MW propagation. [4] (76 Bh, 73 Ma)

9. Explain characteristics of SW radio propagation. [3] (73 Bh,71 Bh) [4] (80 Ch)
10. With a neat diagram, explain the designation of radio waves according to the path they follow during propagation. [4] (70 Bh)
11. Compare the propagation characteristics for different radio bands. [6] (70 Bh)

4.6 Numerical

1. A microwave link is 70 km long. It operates at 2.8 GHz with antenna gain of 20dB at either end. What is the voltage available at the terminals of a receiver with 50 Ohm load impedance if the transmitted power is 2.4W? Assume that it is line of sight communication in case of neglecting atmospheric losses and ignoring problems of fading . Also find out the free space loss for the communication link. [7+2] (81 Ash)
2. A 100 MHz circuit consists of a transmitting and receiving antenna of 30dB and 25dB gains respectively. The power radiated by the transmitting antenan is 120 watts. Find received power in dB at a distance of 0.75 km over a free space. [5] (79 Ash)
3. Find the received power in dB at a distance of 50 km over a free space with 10 GHz frequency consisting of transmitting antenna with 25 dB gain and a receiving gain of 20 dB. The power radiated by transmitted antenna is 150 dB. [6] (79 Ch)
4. A free space LoS microwave link operating at 10 GHz consists of a transmit and receive antenna each having gain of 25 dB. Calculate the path loss of the link nd received power of 100 km distance. Derive Friis transmission equation. [3] (78 Ch)
5. A parabolic reflector antenna having antenna efficiency 75% is designed for 3 GHz resonant frequency with 2.5 dB waveguide loss. Find out the antenna diameter if effective isotropic radiated power (EIRP) is calculated 46 dBW and transmitting power is 500 W. [7] (77 Ch)
|→resonant frequency = 5 GHz, 3 dB loss, EIRP = 50 dBW, power = 600W. [8] (74 Bh)
6. Microwave link is assumed to be free space condition. The antenna gains are each 40 dB, the frequency is 10 GHz and the path length is 90 km. Calculate the transmission path loss and received power of 10 kW. [6] (75 Bh)
7. Find the received power (in dBm) at a distance of 0.5 km over a free space 1 GHz circuit consisting of a transmitting antenna with 25 dB gain and a receiving antenna gain of 20 dB. The power radiated by the transmitting antenna is 150 W. [5] (75 Ba)
8. A parabolic reflector antenna having the antenna efficiency 85% is designed for 3 GHz resonant frequency with 2.5 dB wavelength loss. Find out the antenna diameter if EIRP is calculated 46 dBW and trasmitting power is 500 W. [8] (73 Bh,71 Bh)
9. Assume the reflection take place at a height of 400 km and that the maximum density in the ionosphere corresponds to a 0.8 refractive index at 15 MHz. What will be the range (assume flat earth) for which the MUF is 20 MHz. [5] (70 Ma)

5 Propagation between Antennas

(6 Hours / Marks)

5.1 Free Space propagation

5.1.1 Power density of receiving antenna

1. What is free space propagation? [2] (80 Ash)
2. Derive Friis transmission equation for free space propagation. [7] (80 Ch, 76 Ba)
|→How do you get Friis transmission equation and path loss in case of free space wave propagation?
[4+2] (80 Ash) [4+3] (71 Bh)
3. Derive Friis transmission equation. [2] (76 Bh)
4. Write short notes on: Friis transmission equation [3] (73 Bh)
5. Define free space communication. [2] (75 Ba)

5.1.2 Path loss

1. Derive path loss in case of free space propagation. [2] (76 Bh) [3] (79 Ash) [5] (71 Ma)
|→Derive complete equation including path loss using friis space communication. [8] (75 Ba)
|→Derive the expression for the path loss in case of radio wave propagation. [6] (73 Bh)
2. Write short notes on: Friis equation and path loss in case of free space propagation. [3] (72 Ash)
3. Find out distance of line of sight as: $d = 3.57(\sqrt{h_t} + \sqrt{h_r})$ km. [3] (76 Bh)
4. Calculate free space path loss for 6 GHz frequency, if transmitter and receiver are 10,500 km line of sight apart. [3] (71 Ma)

5.2 Plane earth propagation

5.2.1 Ground Reflection

5.2.2 Effective antenna heights

1. Write short notes on: Effective antenna height. [3] (78 Ch)

5.2.3 The Two Ray

1. Write short notes on: Two-ray propagation model. [3] (81 Ash, 79 Ash)

5.3 Propagation Model, Path Loss

5.4 Fresnel Zones and Knife edge diffraction

1. Define shadow region. [2] (79 Ash)
2. Write short notes on Fresnel knife edge diffraction.
[2] (71 Ma, 70 Ma) [3] (81 Ch, 80 Ch, 79 Ch, 77 Ch, 76 Bh, 72 Ash, 72 Ma) [4] (75 Bh, 74 Bh, 75 Ba, 73 Ma) [5] (70 Bh)
3. Explain knife-edge and rounded surface surface diffraction effects. [6] (79 Ash)

4. Write short notes on Fresnel Zones. [2] (71 Ma) [3] (**78 Ch**)
(Original "*Firesnel*" was assumed to be *Fresnel*)

5.5 Numerical

1. Find out the free space loss (L_{FS}) equation that equals to $22 + 20 \log(\frac{d}{\lambda})$, where d is the distance between T_X and R_X antennas and λ is the wavelength of the frequency being used. [3] (**80 Ch**)
2. What is the maximum power that can be received over a distance of 15.5 km line-of-sight free space with a 3.4 GHz frequency consisting of the transmitting antenna gain of 25 dB and receiving antenna diameter 6.4m with 75% antenna efficiency. Where transmitted power is 250W. [7] (76 Ba)
3. The antenna of a TV transmitter is located at a height of 125m above ground level. Calculate the distance upto which the LOS communication is possible if the height of receiving antenna is to be 9m. [6] (**74 Bh**)
4. Find out the line of sight of distance between the transmitting antenna and receiving antenna. If the transmitting antenna height is 45 m, and the receiving antenna height is 25 m. [Given the radius of Earth is 6,378 km]. [6] (**72 Ash**)
5. A 100 MHz circuit consists of a transmitting and receiving antenna of 30 dB and 25 dB gains respectively. The power radiated by the transmitting antenna is 120 W. Using Friis transmission equation, find the received power at a distance of 0.75 km over a free space. [6] (**72 Ash**)

6 Optical fibres

(11 Hours / Marks)

6.1 Optical Fibre Communication System

1. What is optical fiber? [1] (72 Ma)
2. Explain the block diagram of optical communication system. [3] (**78 Ch**, 71 Ma)
3. Draw the optical fiber communication system. [3] (**74 Bh**)
4. What are the various elements of an optical communication system? Explain each element in brief. [6] (76 Ash) [8] (**73 Bh, 72 Ash**)
5. What properties do you think to be considered when an optical fiber is to be selected for any communication system? [6] (72 Ma)

6.1.1 Merits/Demerits

1. What are the adv and disadv of optical fiber over wireless communication system? [5] (**81 Ch**)
|→ State adv of optical fiber compared to microwave communication link. [4] (**78 Ch**)
|→ Explain adv and disadv of optical fibers over metal wire communication. [5] (**76 Bh**) [6] (**74 Bh**)
|→ Briefly explain the adv of optical fiber over metallised wire. [4] (**79 Ash**) [6] (70 Ma/de)
|→ List out the adv and disadv of optical communication over cable communication. [5] (71 Ma)
2. Explain the block diagram of optical communication system. [3] (**78 Ch**)
3. Where are optical fibers most widely used? [2] (**76 Bh, 70 Bh**)
4. Compare optical fiber communication with cable and radio communication system? Explain each element in brief. [8] (**73 Bh**)

6.2 Types of optical fibre and its Structural differences

1. Explain the contribution of microscope and macroscopic fiber bends towards the bending losses in optical fiber. [7] (**77 Ch**)
2. Explain the construction and propagation mechanism and application of different types of optical fibers. [5] (**76 Bh**) [8] (73 Ma) [10] (**71 Bh**)
3. Explain the construction and types of optical fiber with necessary diagrams. [5] (76 Ba)
|→ Compare the configuration of different types of fiber. [4] (**72 Ash**)
4. What are the various feature of graded index fiber? [3] (76 Ba)
5. Write short notes on: Multimode graded index fiber. [4] (73 Ma)
6. Explain refractive index profile and ray transmission in a multimode graded index fiber. [5] (76 Ba)
7. List adv & disadv of multimode fiber over the single mode fiber? [3] (70 Ma) [5] (**81 Ash**, 72 Ma)
|→ Mention the advantages of single mode fiber. [2] (**72 Ash**)
8. Discuss the structure and refractive index profile of single mode and multi-mode step index fibres with related mathematical equations. [7] (79 Ash)

6.3 Light propagation characteristics

1. Explain the working principle of optical fiber communication system. [3] (81 Ch)
2. Write short notes on: Super refraction. [4] (74 Bh)
3. Explain the principle of light propagation through a fiber. [2] (79 Ch)
|→with necessary diagram. [5] (80 Ch)
|→Describe with neat diagram the basic principle of total internal reflection that enables the fiber to work as a “light conduit”. [6] (75 Bh)
4. What is an acceptance angle? [2] (81 Ch,78 Ch,70 Bh)
5. Derive an expression to calculate the acceptance angle. [7] (70 Bh)

6.4 NA in optical fibre

1. Derive the expression for numerical aperture of a stepped index optical fiber. [5] (81 Ch)
|→Derive the relationship between numerical aperture and acceptance angle. [6] (80 Ch)
|→Derive the expression for NA with neat diagram. [5] (78 Ch)
|→Derive expression of NA of a stepped index optical fiber. [7] (70 Ma)
2. Define numerical aperture. [2] (77 Ch)
|→with necessary figures and mathematical expression. [2] (79 Ch)
3. Define acceptance angle with necessary figures and mathematical expression. [2] (79 Ch)
4. Write short notes on: NA. [3] (81 Ash, 71 Ma) [4] (74 Bh)
|→Write short notes on: NA for meridional rays in optical fiber. [3] (72 Ash)
5. Describe the following term: Dispersion. [3] (71 Ma)

6.5 Losses

1. Write short notes on: Optical fiber loss. [3] (79 Ch)
|→Explain losses in optical fibre. [4] (79 Ash)
2. Discuss loss or signal attenuation in an optical fiber with respect to absorption, scattering and bending losses. [8] (75 Ba)
3. What are the different types of losses you can visualize in optical fiber communication? [4] (81 Ash, 72 Ma)

6.6 Light Source and Photo Detector

1. Write short notes on: Light source and photo detector. [3] (80 Ch)
|→on: Optical source and Optical detector. [3] (80 Ash) [4] (77 Ch,75 Bh, 75 Ba)
2. Write short notes on: Critical frequency. [4] (76 Ba)
3. List different optical sources & explain the losses in optical fiber briefly. [6] (77 Ch) [2+5] (75 Bh)

7 Don't know where this fits

Write short notes on: Splicing. [3] (79 Ch)